Vacuum fluctuations are, for electromagnetic and gravitational waves, what "claustrophobic degeneracy motions" are for electrons.

Recall (Chapter 4) that if one confines an electron to a small region of space, then no matter how hard one tries to slow it to a stop, the laws of quantum mechanics force the electron to continue moving randomly, unpredictably. This is the claustrophobic degeneracy motion that produces the pressure by which white-dwarf stars support themselves against their own gravitational squeeze.

Similarly, if one tries to remove all electromagnetic or gravitational oscillations from some region of space, one will never succeed. The laws of quantum mechanics insist that there always remain some random, unpredictable oscillations, that is, some random, unpredictable electromagnetic and gravitational waves. These are the vacuum fluctuations that (according to Zel'dovich) will "tickle" a spinning metal sphere or black hole and cause it to radiate.

These vacuum fluctuations cannot be stopped by removing their energy, because they contain, on average, no energy at all. At some locations and some moments of time they have positive energy that has been "borrowed" from other locations, and those other locations, as a result, have negative energy. Just as banks will not let customers maintain negative bank balances for long, so the laws of physics force the regions of negative energy to quickly suck energy out of their positive-energy neighbors, thereby restoring themselves to a zero or positive balance. This continual, random, borrowing and returning of energy is what drives the vacuum fluctuations.

Just as an electron's degeneracy motions become more vigorous when one confines the electron to a smaller and smaller region (Chapter 4), so also the vacuum fluctuations of electromagnetic and gravitational waves are more vigorous in small regions than in large, that is, more vigorous for small wavelengths than for large. This, as we shall see in Chapter 13, has profound consequences for the nature of the singularities at the centers of black holes.

Electromagnetic vacuum fluctuations are well understood and are a common feature of everyday physics. For example, they play a key role in the operation of a fluorescent light tube. An electrical discharge excites mercury vapor atoms in the tube, and then random electromagnetic vacuum fluctuations tickle each excited atom, causing it, at some random time, to emit some of its excitation energy as an electromagnetic wave (a photon).*